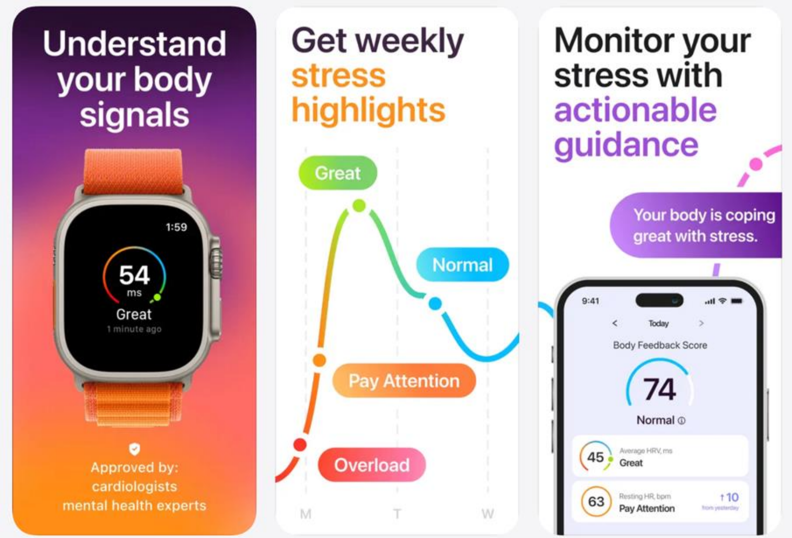
Course Project

Emotion Recognition from Photoplethysmography (PPG) Signals

CSC 491/591
Ubiquitous Computing and Mobile Health

Background

PPG signals reflect cardiovascular activity and can be captured by low-cost wearable devices. Features derived from PPG — such as heart rate and HRV — are known indicators of emotional and physiological arousal.



Goal of the Project

1. Process raw PPG Data from a PPG-based emotion recognition dataset.

2. Build Emotion Classification Models on the dataset.

Dataset Introduciton

Wearable Stress and Affect Detection (WESAD) is a dataset designed for stress and affect recognition. It contains physiological and motion data collected from 15 subjects under different experimentally induced emotional conditions.

Dataset Download

Dataset Introduciton: File Structure & Access

```
WESAD/
 ├── wesad_readme.pdf
 ├── S2/
 |    ├── S2.pkl
 |    ├── S2_E4_Data.zip
 |    ├── S2_respiban.txt
 |    ├── S2_quest.csv
 |    └── S2_readme.txt
 ├── S3/
 ├── S4/
 |   ...
 └── S17/
```

Tips:

1. Read **wesad_readme.pdf** first

It contains all field definitions you need before touching the data.

2. Use SX.pkl directly

It is the simplest and recommended access method; all modalities are pre-synchronised and ready to load (see Section II.3 *Synchronised Data* in the readme)

3. We only need two fields from SX.pkl: PPG signal (BVP) and emotion labels

Project Tasks & Report Guidelines

Section 1：Data Processing（60 pts）

What to Do:

Process the raw wrist PPG signal from WESAD.

What to Write:

Detail every processing step and justify your choices. After completing all preprocessing steps, include a **visualization** of at least one processed PPG segment.

*Visualization example of **raw** ppg segments*

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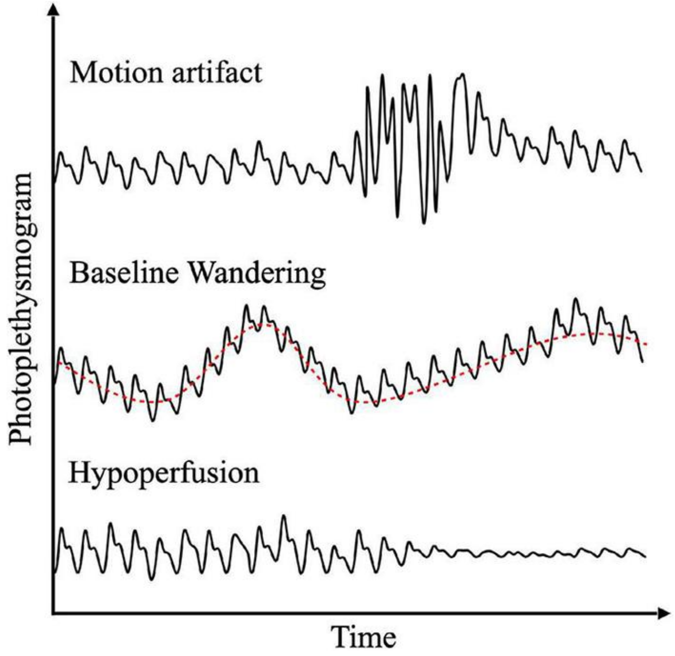
Data Processing: Filtering

Why Filtering?

Baseline Wander Low-frequency drift caused by breathing motion and body movement. Shifts the signal baseline up and down, making peak detection unreliable.

High-Frequency Noise Electrical interference and sensor noise introduce high-frequency components that have nothing to do with cardiac activity.

Motion Artifacts Physical movement of the wrist causes the sensor to shift, producing large amplitude distortions that can be mistaken for heartbeats.



Data Processing: Filtering

Approach: Butterworth Bandpass Filter

A Butterworth filter is chosen for its maximally flat frequency response in the passband — it preserves the cardiac signal without distorting waveform morphology.

Key Parameters:

Low cutoff: 0.5 Hz

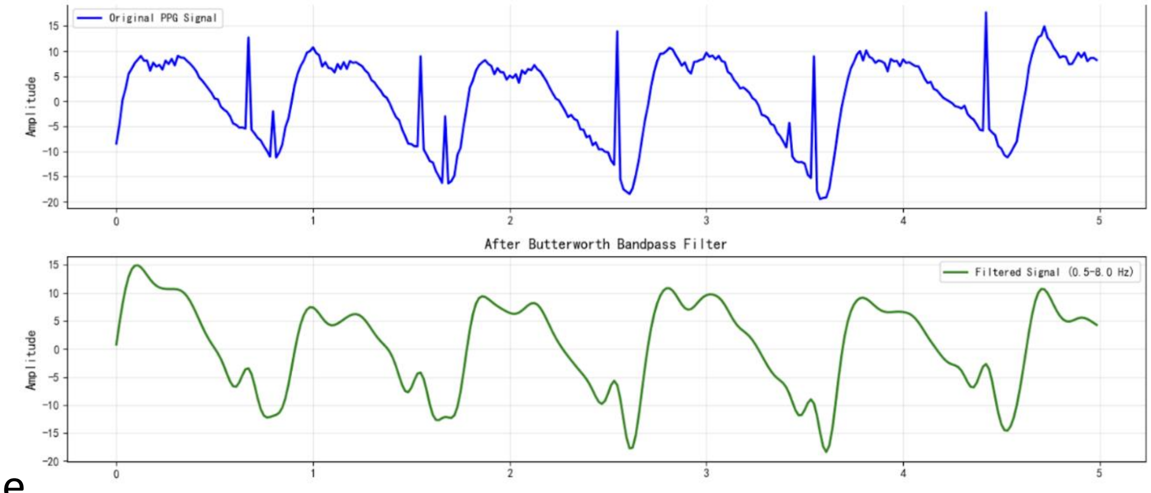
Remove baseline wander (< 0.5 Hz)

High cutoff: 4 Hz

Remove high-frequency noise; 4 Hz corresponds to 240 BPM upper limit

Order: 4th

Balance between sharpness and phase distortion



Data Processing: Segmentation

The raw WESAD PPG recording is one long continuous signal spanning the entire experiment session, which includes multiple emotional conditions.

Through segmentation, we will solve:

Mixed labels: A single continuous recording contains multiple emotional states. Without segmentation, each sample has no single clear label.

No fixed-size input: ML/DL models require fixed-size input vectors. A continuous signal of varying length cannot be directly used.

Insufficient training samples: 15 subjects is a small number. Segmentation with overlapping windows multiplies the number of usable samples for training.

Data Processing: Windowlization

Key Parameters:

Window size

Window step

Project Tasks & Report Guidelines

Section 2: Methodology (30 pts)

What to Do:

Implement **at least one emotion classifier** (SVM, Random Forest, 1D CNN, LSTM, …).

What to Write:

Provide key implementation details — for classical models, list key hyperparameters; for deep learning models, include a diagram illustrating the model architecture with layer types, dimensions, and key parameters

Emotion Classification Models

Approach 1 — End-to-End

Raw PPG windows fed directly into the model; the model learns its own representations.

Approach 2 — Feature-Based

Extract hand-crafted features from PPG first, then train a classifier.
Features: HR, HRV (RMSSD, SDNN, pNN50), LF/HF ratio, IBI, …

You are encouraged to implement models from both approaches and compare their performance.

Report Requirements

Section 3: Experiments & Analysis (10 pts)

Present and compare classification results across all implemented models using appropriate metrics, such as Accuracy, F1 Score, and Confusion Matrix.

Try your best to give us some insights.

Tips & References

Tips:

1. **Carefully read the reference papers below first.**

2. **Train/Test Split by Subject**

When splitting data, split by subject rather than by windows. This ensures the model is evaluated on unseen subjects, which better reflects real-world generalization ability and avoids data leakage.

References

WESAD Paper

Feature Augmented Hybrid CNN for Stress Recognition Using Wrist-based Photoplethysmography Sensor

Use any programming language you are comfortable with

Choose Sole or Team up. At most two students per group.

Group Registration by **March 9th**

Submit a project report by **April 27th**

Team Match

For any questions regarding the course project, feel free to email me at kshen5@ncsu.edu